Scope

This SOP applies to **all escalated support work** after a ticket leaves the general support queue and requires product/engineering/service investigation or action, including:

- **Incidents** (production-impacting events)
- **Non-incident defects/bugs**
- **By-design / education / environment** outcomes

Objectives

- Make prioritization of defects and incidents automatic and clear so the Product team can respond to them in a clear, consistent and non-disruptive way
- Restore/mitigate customer impact quickly (incident or non-incident)
- Provide predictable updates without meetings or status-chasing
- Make the **integration** between **incident response** and **defect/bug execution** explicit
- Ensure ownership never goes dark: every escalated ticket always has an owner
- Capture decisions, links, and outcomes so Support and customers can validate closure cleanly

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Systems and Source of Truth

- Tickets exist in **HubSpot** and **ADO/VSO/TFS** via synchronization integrations.
- **HubSpot** is the source of truth for customer-facing status and communication.
- **ADO** is the execution system for engineering work.

Definitions

Defect (Bug)
A confirmed product fault requiring an engineering change (code/config/data fix or prevention work) and tracked to completion.

Product Incident/Service Incident
An unplanned event that degrades, interrupts, or risks the confidentiality, integrity, or availability

of a production service or business-critical workflow requiring immediate investigation, mitigation, or communication.

Outage
A Service Incident resulting in unavailability of a service or critical customer journey for a defined population.

Degradation
A Service Incident where the service is available but fails to meet performance, reliability, or functional expectations (e.g., elevated error rate, SLO breach, partial feature failure).

Data/Integrity Incident
A Service Incident where data is incorrect, missing, duplicated, unauthorizedly altered, or exposed; includes security and privacy events.

Customer Impact
Any measurable negative effect to customers (inability to complete a journey, incorrect results, delayed processing, data exposure, or loss of trust), described by impacted population and duration.

Mitigation
A change that reduces or stops customer impact without necessarily removing the underlying cause (e.g., rollback, feature flag off, config change, traffic shift).

Resolution
The underlying cause is fixed and verified (e.g., patch deployed and validated). Resolution may occur after mitigation.

Workaround
A customer-usable alternative path that allows the workflow to continue. Workarounds affect urgency/severity but do not eliminate the need for a proper fix if the defect remains.

Severity (S1–S4)
Incident severity used for incident response handling.

Priority (P0–P3)
Defect/support priority used for non-incident execution and update SLAs. Priority is separate from incident severity.

Next Update By
A required checkpoint field used to prevent “no ETA” status-chasing. ETAs may be unknown, but **Next Update By** cannot be empty.

RCA (Root Cause Analysis)
A required document that captures the incident summary, cause, impact, timeline, and

prevention follow-up actions.
RCA Trigger: An incident requires RCA if it meets defined severity or impact thresholds (S1/S2, security/privacy, material data/financial impact, SLA breach, recurrence, or significant manual intervention).

Definition of Ready (DoR)
The minimum information required for engineering to pull and progress a ticket without avoidable back-and-forth.

ADO (VSO, VSTS, TFSO)
Azure DevOps. Formerly called Visual Studio Online (VSO), Visual Studio Team Services (VSTS), and Team Foundation Server/Services.

Ticket Types

Support Ticket (ST)
The originating ticket handled by frontline support. Support Tickets are out of scope for the Product and Services team unless escalated.

Technical Escalation Ticket (TET)
A child ticket created from a Support Ticket when frontline support escalates work to the Product and Services team for investigation or action. A TET is the primary in-scope work record for this SOP. A TET may end as a confirmed defect, incident-linked issue, by-design outcome, education outcome, environment issue, or duplicate.

Service Incident Ticket (SIT)
A ticket created when any type of qualifying service incident is occurring.

Roles

Frontline Support Personnel
Point of first contact for support requests coming from customers via tickets, phone, email or other direct contact methods.

Technical Triage Support Lead (TSTL)
Responsible for ingestion and accuracy of TET tickets to the product team. They ensure that the TET fields are accurate so that TET tickets and SIT tickets are prioritized accordingly.

Product Manager (PM)
Responsible for ensuring that the product delivery cycle remains on course and that defects and incidents are handled in accordance with our defined processes and SOPs.

Developer
A software developer or, for the purposes of this document, any technical role that may be

handling technical support tickets.

Policies

Ownership

Every escalated ticket must always have an Owner assigned. Owner must never be blank.

- Owner may be a person or a team queue.
- Once someone owns a ticket they own it until completion or explicit transfer to another person or queue

Communication Policy

- Asynchronous communication. No status meetings by default.
- If someone asks “status?”, the most up to date answer is in the HubSpot ticket/record.
- Escalate to synchronous only if:
  - P0 is blocked and coordination is required
  - **Next Update By** is repeatedly missed
  - Multi-team decisions require live alignment

“No ETA” Policy

- ETAs may be **Unknown**
- But **Next Update By cannot be empty**
- Every update must state what will be learned/done by the next update
- Written communications are considered customer-safe unless specifically noted

Engineering Update Expectation (Business Days)

For any escalated ticket a developer owns: provide a **meaningful update by the ticket's Next Update By deadline** while open and active/blocked.

A meaningful update includes at least one of:

- What changed since last update
- Current hypothesis/findings
- What is blocking and what’s needed
- What will be done next + next checkpoint

High Level Workflows

Normal (Non-Incident Flow)

Step by Step

1. Customer contacts support (Support Ticket Created)
   - support gathers context and confirms the issue needs technical escalation (or routes elsewhere)
2. Support creates a Technical Escalation Ticket (TET)
   - parented from the original ticket
   - this process is easy and minimally impactful for frontline support personnel
     - support should be able to click from the original ticket they are in and be prompted with the form fields for the TET
     - they should not have to remember to search for a form and fill it out as an out-of-band process
3. Tech Support Triage Lead (TSTL) reviews and adjusts ticket fields
   - using automatic scoring system
   - Outputs:
     - defect priority (P0-P3)
   - TSTL handles/assigns any TET tickets they are able to
4. TSTL routes the TET based on what is needed next
   - TETs that require engineering work are escalated to the dev team
   - TETs that do not require engineering work may be handled by the TSTL or routed elsewhere
5. TSTL, the dev team, and PM are responsible for ensuring tickets are handled within their SLAs
   - TSTL reviews `New` tickets
   - developers review `Ready for Dev` and engineering-owned open tickets
   - PM ensures the overall workflow is functioning and SLA expectations are being met
   - assign tickets to individuals for work
   - set statuses of tickets based on what they need
   - set Next Update By field
   - close out tickets that are completed or will not be worked on
   - SLAs will be defined for different priorities elsewhere in this document
     - updates, deferral dates, etc. will be expected based on the SLAs

6. Tech Triage & Frontline Support are able to view up to date statuses of the tickets and manage most customer facing communication
    - ticket update SLAs are important in order to reduce the need for synchronous communication and meetings in order to get the status of defects and incidents
7. Closure
    - fix shipped and verified

Incident-Interrupt Flow (When an Incident Ticket Is Created)

Step by Step

Steps 1 and 2 remain the same.
3. The incident is triggered by the IncidentScore. This is triggered based on the form being filled out. The incident may be triggered upon initial creation by support or by the Tech Triage. It is automated.

- It must be obvious to support and to Tech Triage that it has triggered an incident
- Developers and PM will receive alerts that an incident is occurring
- Linked to the parent ticket(s)
- Incident creation is forward-moving only. Once an incident is created, it is never silently removed or "uncreated" by later field changes.
- If an incident was created in error or later investigation shows incident handling is not required, the incident may be closed, but the TET remains.
  4. Update cadence and SLAs become incident-driven
- these will be different than SLAs for non-incident defects
- devs update the incident tickets
  5. Mitigate first, fix later
- incident can move to Mitigated/Monitoring while permanent defect is still in progress
  6. RCA required if S1/S2/security incident
  7. Closure
  - Incident is closed when the impact is mitigated + communications cadence is satisfied + RCA is completed (if required)

Non-Incidents

Roles

Technical Support Triage Lead (TSTL)

- Owns the Product and Service Support Queue
- Owns customer communication plan and ensures HubSpot stays customer-safe and current
- Ensures required intake fields are correct
- Coordinates routing and links (companies/contacts/related tickets/incidents)
- owns the primary support queue
- escalates tickets to the Product and Service Support Queue

Non-Incident TET Statuses

Open

1. **New**
2. **Ready for Dev**
3. **Investigation (Eng)** *(notes required)*
4. **In Progress (Fix/Change)** *(notes required)*
5. **Waiting on Customer** *(notes required)*
6. **Fix Deployed / Monitoring** *(notes required)*
7. **Resolved (Pending Closure)** *(notes required)*

Closed

1. **Closed (Fixed)**
2. **Closed (No Defect / Education)**
3. **Closed (Declined / By Design)**
4. **Merged / Duplicate**

Priority-Based SLAs (Non-Incident)

These SLAs are business-time expectations for non-incident TETs. Incident SLAs use a separate elapsed-time model.

Priority determines three timers:

- maximum time a ticket may remain in `New`
- maximum time a ticket may remain in `Ready for Dev`
- maximum time until the next required update while the ticket remains open

Priority Determination

Priority is driven by support form scoring inputs (impact/reach/frequency/workaround/time-criticality, etc.) and is used to determine response and update expectations.

Scoring Model

The support form calculates three separate outputs:

- **BugScore** for non-incident defect priority
- **IncidentScore** for whether an incident ticket must be created
- **SeverityScore** for incident severity handling

Priority and severity are related but separate:

- **Priority (P0-P3)** governs non-incident execution and update expectations
- **Severity (S1-S4)** governs incident response handling

Scoring Inputs

Bug/Defect scoring inputs

- **Impact** = 1-5
- **Reach** = 1-5
- **Frequency** = 1-5
- **Workaround** = 0, 1, or 3
- **TimeCriticality** = 0-3
- **CustomerImportance** = 0-3
- **Sentiment** = 0-2

Incident scoring inputs

- **ProductionImpact** = Yes/No
- **ImminentRisk** = 0, 2, or 6
  - `Unlikely / No chance = 0`
  - `Maybe / Possible = 2`
  - `Likely / Certain = 6`
- **Availability** = Yes/No
- **Performance** = Yes/No
- **DataIntegrity** = Yes/No
- **SecurityPrivacy** = Yes/No
- **BillingFinancial** = Yes/No
- **Operational** = Yes/No

BugScore -> Priority

```
BugScore =
  (Impact * 2)
+ Reach
+ Frequency
+ Workaround
+ TimeCriticality
+ CustomerImportance
+ Sentiment
```

Why these inputs are weighted this way:

- impact is double weighted because severity of workflow disruption matters most
- reach and frequency reflect how widespread and repeatable the issue is
- workaround and time criticality capture urgency
- customer importance and sentiment capture commercial and relationship risk

Priority thresholds:

- **P0** if `BugScore >= 26`
- **P0** if `Impact = 5` and `Reach >= 3`
- **P1** if `BugScore = 20-25`
- **P2** if `BugScore = 14-19`
- **P3** if `BugScore <= 13`

IncidentScore -> Incident Trigger

```
IncidentScore =
  (ProductionImpact ? 10 : 0)
+ ImminentRisk
+ (SecurityPrivacy ? 10 : 0)
+ (DataIntegrity ? 6 : 0)
+ (BillingFinancial ? 6 : 0)
+ (Impact >= 4 ? 3 : 0)
+ (Reach >= 3 ? 3 : 0)
+ Workaround
```

Incident creation rule:

- Create an incident ticket when `IncidentScore >= 10`

Operational interpretation:

- a current production impact always triggers an incident
- a security/privacy impact always triggers an incident
- otherwise, enough combined risk and customer-impact factors can still trigger an incident

SeverityScore -> Severity

```
SeverityScore =
  (ProductionImpact ? 12 : 0)
+ ImminentRisk
+ (Availability ? 8 : 0)
+ (Performance ? 5 : 0)
+ (DataIntegrity ? 8 : 0)
+ (SecurityPrivacy ? 12 : 0)
+ (BillingFinancial ? 8 : 0)
+ (Operational ? 4 : 0)
+ (Impact * 3)
+ (Reach * 2)
+ Workaround
```

Severity thresholds:

- **S1** if `SeverityScore >= 40`
- **S2** if `SeverityScore = 30-39`
- **S3** if `SeverityScore = 20-29`
- **S4** if `SeverityScore < 20`

Severity levels determine incident handling expectations. They do not replace non-incident priority.

SLA Timer Table

Priority	Max Time in `New`	Max Time in `Ready for Dev`	Max Time Until `Next Update By`
P0	4 business hours	4 business hours	24 business hours
P1	8 business hours	1 business day	48 business hours
P2	2 business days	5 business days	5 business days
P3	3 business days	10 business days	5 business days

SLA Notes

- `New` is the TSTL initial triage timer.

- `Ready for Dev` is the timer for developer pickup from the prioritized queue.
- `Next Update By` is the maximum time until the next meaningful update while the ticket remains open.
- `Next Update By` should be reset on each update until the ticket reaches a closed state.
- Reminder/nagging behavior should be driven by these timers so communication stays current without manual status-chasing.

Detailed Non-Incident Procedure

0) Escalation Intake (From General Support)

Trigger: Ticket is escalated into Product & Service Support.

Actions

- Technical Escalation Ticket (TET) created
- Ensure ticket has:
  - impacted company/contact(s) linked
  - clear description (expected vs actual + repro)
  - customer impact described in plain language
- Complete scoring inputs to the best ability of the support personnel

Definition of Done

- TET created
- Ticket priority is calculated
- Incident trigger and severity are calculated

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1) Incident Short-Circuit Check (Required and Continuous)

Rule

If Incident criteria are met at any time, stop the non-incident procedure and the Service Incident Procedure will be triggered

Minimum actions when short-circuiting

- HubSpot and ADO incidents are created
- Originating HubSpot ticket(s) are linked to incident tickets in HubSpot and ADO
- Continue under Incident SOP roles/cadence/status

- Do not wait for perfect intake completeness when current production impact or security/privacy impact is already known

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2) Triage and Definition of Ready (DoR)

Goal: Gather only what's necessary to make prioritization and handling of tickets quick and clear

DoR minimums (must be present or explicitly "unknown")

- Expected vs actual behavior
- Repro steps (or "cannot reproduce" details)
- Environment details (tenant/org, user, permissions, feature flags, integration context)
- Time window + identifiers (timestamps, job IDs, record IDs)
- Evidence where possible (screenshots, logs, error messages)

Actions

- `New` means the TET is in the P&S queue and awaiting initial triage by the TSTL
- TSTL ensures DoR minimums are met
- If customer info is required, move ticket to "Waiting on Customer" and specify exactly what's needed
- If the TET is sufficiently understood and ready for engineering work, move it to `Ready for Dev`
- P0 defects will move to the top of the queue and displace any other non-P0 tickets that have been picked up or committed to
    - any displaced tickets need to have a status update as to the reason

Definition of Done

- Ticket is assigned to the P&S queue
- Ticket is moved out of `New`
- Tickets that don't require developer action can be handled by the TSTL or assigned elsewhere by the TSTL

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3) Developer Pickup and Ingestion

Goal: Outstanding tickets are reviewed and updated at least once a day

Actions

- Development team runs through the outstanding tickets once a day
- Pull tickets from `Ready for Dev` in priority order
- Change ticket statuses from `Ready for Dev` to another active working state when work begins
- Provide next update times and update notes
- The highest priority Tickets are pulled by developers for further investigation and action

Definition of Done

- no tickets remain in `New` after initial triage
- tickets are assigned to individual developers where it makes sense

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4) Engineering Investigation → Work → Resolution (with SLA Updates)

Goal: Investigate, create/track any needed work items, resolve the issue (or return it with actionable next steps), and keep regular updates within SLA.

Actions

- Developer takes the ticket (becomes Owner) and remains accountable end-to-end
- Investigate and document findings in ADO ticket (linked to HubSpot ticket)
- Create/link new ADO work item(s) as appropriate (task/bug/etc.)
- Drive the work to resolution (fix/change/answer) or return with exactly what's needed next to proceed
- Provide timely, regular updates in HubSpot within SLAs, including:
  - customer-safe summary
  - current state
  - Next Update By

Definition of Done

- Tasks/Work Items/etc. created and tracked
- Issue is resolved or handback includes actionable next steps
- Updates were provided within SLA and Next Update By is set appropriately

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5) Closure (Non-Incident)

When closing an escalated ticket, ensure the record includes:

- resolution summary (what was fixed or confirmed)
- tests/verification notes as appropriate
- release/deployment reference if applicable
- how support can validate with the customer (clear steps)
- links to ADO work items

Definition of Done

- Ticket is closed with the correct closure reason/status
- Customer-facing narrative is complete and verifiable
- Any follow-up work (docs/runbooks/prevention) is captured as linked work items

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Incidents

Incident Roles

At the outset of the incident, people will declare themselves into these roles in the HubSpot Incident ticket

Incident Commander (IC)

- Owns incident progression, updates cadence, and decisions
- Ensures **Next Update By** is always set
- Coordinates responders, but does not need to write code
- Captures key timeline notes (can be the IC in small teams)

Primary Responder (Engineering)

- Drives technical diagnosis/mitigation
- Updates ADO Incident fields (customer-safe summary + next update)

Support Lead (Customer Comms Owner)

- Communicates externally based on HubSpot Incident record
- Ensures impacted accounts are linked + informed

Severity Levels (S1–S4) definitions

These Severity Levels are calculated from fields in the P&S Support Form

- **S1:** widespread outage/blocker for many customers OR security/privacy OR major data integrity/billing impact
- **S2:** major workflow broken for multiple customers/orgs; no acceptable workaround
- **S3:** limited segment impact; workaround exists
- **S4:** minimal impact; informational; often no RCA

> Defect priority (P0–P3) is separate from incident severity (S1–S4). They are both calculated based on many of the same inputs but the handling of each are unique from one another.

Incident Status Workflow (HubSpot)

Open

1. **Suspected / Triage** *(notes required)*
2. **Active Response**
3. **Mitigated / Monitoring** *(notes required)*
4. **Resolved (Pending RCA)** *(notes required)*

Closed

1. **Closed (RCA Complete)** *(notes required)*
2. **Closed (No RCA Required)** *(notes required)*
3. **Merged / Duplicate** *(notes required)*

Communication policy (to eliminate meetings)

- No “status meetings” by default
- Stakeholders subscribe to Teams incident feed
- If someone asks “status?”, the answer is the HubSpot Incident record
- Only escalate to synchronous if:
  - Severity S1/S2 and mitigation is blocked
  - Next Update By is missed repeatedly
  - Multi-team coordination requires live decision-making

Incident SLAs

Incident SLAs are severity-based and use elapsed time, except where explicitly stated otherwise.

Severity	Max Time Until `Next Update By`
S1	1 hour
S2	2 hours
S3	4 hours
S4	1 business day

Procedure

1) Declare Incident

Trigger: Incident criteria met

> This is calculated but the logic is that an Incident is triggered if there is current production impact OR security or privacy impact OR enough other impact factors are involved (customer impact, billing/financial, reach, workaround availability, etc.)

Actions

- Create HubSpot Incident ticket (happens automatically)
- Create ADO Incident parent work item (happens via integration)
  - Link HubSpot Incident ↔ ADO Incident
- Associate impacted companies/contacts + related tech tickets
  - See Incident Roles above
- Incident creation may happen on initial intake or later as ticket fields are updated
- Once created, the incident record remains even if later scoring or investigation reduces the apparent impact
- Assign **Incident Commander**
- Set **Severity** + **Impact dimensions**
  - this occurs in the support ticket
- Set **Next Update By** *(required)*

Definition of Done

- Incident is visible in dashboards + Teams channel feed

- Incident Roles + update cadence are set
- Severity reflects the best current known facts and may be recalculated as incident fields are updated

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2) Active Response

Goal: mitigate impact, identify containment, stabilize.

Actions

- Engineering focuses on mitigation (rollback/flag/config/capacity)
- Support Lead updates customers using HubSpot incident fields
- IC ensures updates land on schedule (no chasing)

Required engineer update fields (ADO Incident)

- Current Status
- Latest Update (customer-safe)
- Next Update By
- (Optional) ETA Mitigation / ETA Resolution (or "Unknown")

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3) Mitigated / Monitoring

Goal: ensure impact stays reduced/ended; watch for regression.

Actions

- Record mitigation method and monitoring signals
- Decide monitoring window (e.g., 30–60 minutes or "one full batch run")
- Update HubSpot incident to Mitigated/Monitoring with required notes:
  - what mitigation was applied
  - what "stable" looks like
  - monitoring duration and signals

Definition of Done

- User/Customer impact is initially removed/reduced and it is in a monitoring state
- Metrics stable
- Support has customer-safe phrasing for "mitigated"

4) Resolved (Pending RCA)

Goal: impact ended; transition into learning/prevention.

Actions

- Confirm resolution criteria met
- Confirm all customer comms done or scheduled
- Create RCA document/work item and link it
- Create follow-up ADO tasks/bugs for prevention (owned and prioritized)

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5) Close Incident

- If RCA required: close as **Closed (RCA Complete)**
- If not: close as **Closed (No RCA Required)**
- Ensure the final HubSpot incident has:
    - summary
    - start/end times
    - impacted customers (if applicable)
    - links to ADO incident + defects + RCA

Appendix A — Engineering Update Checklist for Support

This appendix is a quick-reference checklist only. The governing rules for update timing and `Next Update By` are defined in the main SOP above.

When progress occurs, or when work is blocked, Engineering provides:

- current internal status
- plain-language summary of findings
- what changed / what's next
- rough timing if known; or explicitly "unknown" + **Next Update By**